



ACE-SNN: Algorithm-Hardware Co-design of Energy-Efficient & Low-Latency Deep Spiking Neural Networks for 3D Image Recognition

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High-quality 3D image recognition is an important component of many vision and robotics systems. However, the accurate processing of these images requires the use of compute-expensive 3D Convolutional Neural Networks (CNNs). To address this challenge, we propose the use of Spiking Neural Networks (SNNs) that are generated from iso-architecture CNNs and trained with quantization-aware gradient descent to optimize their weights, membrane leak, and firing thresholds. During both training and inference, the analog pixel values of a 3D image are directly applied to the input layer of the SNN without the need to convert to a spike-train. This significantly reduces the training and inference latency and results in high degree of activation sparsity, which yields significant improvements in computational efficiency. However, this introduces energy-hungry digital multiplications in the first layer of our models, which we propose to mitigate using a processing-in-memory (PIM) architecture. To evaluate our proposal, we propose a 3D and a 3D/2D hybrid SNN-compatible convolutional architecture and choose hyperspectral imaging (HSI) as an application for 3D image recognition. We achieve overall test accuracy of 98.68, 99.50, and 97.95% with 5 time steps (inference latency) and 6-bit weight quantization on the Indian Pines, Pavia University, and Salinas Scene datasets, respectively. In particular, our models implemented using standard digital hardware achieved accuracies similar to state-of-the-art (SOTA) with $\sim 560.6\times$ and $\sim 44.8\times$ less average energy than an iso-architecture full-precision and 6-bit quantized CNN, respectively. Adopting the PIM architecture in the first layer, further improves the average energy, delay, and energy-delay-product (EDP) by 30, 7, and 38%, respectively.

Keywords: hyperspectral images, spiking neural networks, quantization-aware, gradient descent, processing-in-memory

1. INTRODUCTION

3D image classification is an important problem, with applications ranging from autonomous drones to augmented reality. 3D content creation has been gaining momentum in the recent past and the amount of information in the form of 3D input data becoming publicly available is steadily increasing. In particular, hyperspectral imaging (HSI), which extracts rich spatial-spectral information about the ground surface, has shown immense promise in remote sensing